

Postdoctoral Fellow, Harvard / *Systems neuroscience, connectomics, modeling* /
Creator of CeDNe, a graph-based platform for multi-omic brain data

CONTACT INFORMATION	Postdoctoral Fellow at Zhang Lab Website: https://sahilmoza.com Harvard University sahil.moza@gmail.com , sahilmoza@fas.harvard.edu
RESEARCH PROGRAM	• Brain-wide mechanisms of learning and behavioral flexibility in <i>C. elegans</i>. • Computational frameworks linking connectome structure to neural dynamics and behavior. • Research software for multimodal neural data integration, simulation, and optimization.
SELECTED ONGOING RESEARCH	Co-leading a multi-institutional project on whole-brain learning dynamics in <i>C. elegans</i> ; context-gated reorganization of neural activity. <i>Manuscript in review.</i> (2023-) CeDNe : Connectome-driven framework for integrating multimodal data with simulation and optimization workflows. <i>Preprint posted.</i> (2024-)
EDUCATION & TRAINING	Postdoctoral Research Dept of Organismic & Evolutionary Biology Sep 2022- Harvard University , Cambridge, USA Mentor: Yun Zhang Postdoctoral Research Boston Children's Hospital Jul 2021-Sep 2022 Harvard Medical School , Boston, USA Scientist , EBRAINS, Human Brain Project Oct 2020 - July 2021 KTH, Stockholm Ph.D Neuroscience/Systems Biology July 2020 National Centre for Biological Sciences Bangalore, India Mentor: Upinder S. Bhalla M.E. Computational and Systems Biology Aug 2012 Jawaharlal Nehru University New-Delhi, India B.E. Biotechnology Jul 2010 Panjab University Chandigarh, India
PEER-REVIEWED RESEARCH ARTICLES	Bhatia, A*, Moza, S* , Bhalla, US, "Precise excitation-inhibition balance controls gain and timing in hippocampus.", <i>eLife</i> , Apr 2019 (*Equal contribution; co-first author) Faculty Opinions (Exceptional) Recommendation. In Faculty Opinions, 04 May 2020 HarshaRani, GV, Moza, S , Ramakrishnan, N, Bhalla, US, "SWITCHES: Searchable Web Interface for Topologies of CHEmical Switches.", <i>Bioinformatics</i> , Jan 2021. Resource website: http://SWITCHES.ncbs.res.in
MANUSCRIPTS UNDER REVISION	Liang*, J, Moon*, S, Moza*, S , Lee*, HJ et al., "Aversive Learning Induces Context-Gated Global Reorganization of Neural Dynamics in Caenorhabditis elegans", <i>under revision at Nature</i> ; preprint: <i>bioRxiv</i> , Nov 2025 (*Equal contribution; co-first author)
PREPRINTS	Moza, S , Zhang, Y, "CeDNe: A multi-scale computational framework for modeling structure-function relationships in the C. elegans nervous system", <i>bioRxiv</i> , Nov 2025 (lead author; framework design and implementation)

	Moza, S, Bhalla, US, “Different dimensions of robustness—noise, topology and rates—are nearly independent in chemical switches.”, <i>bioRxiv</i>, Aug 2020 Moon, S et al. (including Moza, S), “Network modularity reveals context and state-dependent reorganization of time-varying functional connectivity in single-cell resolved neural activity recordings”, <i>bioRxiv</i>, Nov 2025	
RESEARCH SOFTWARE AND INFRASTRUCTURE	CeDNe — graph-based framework for multimodal connectome modeling, simulation, and optimization Jun 2025 NeuroRD-SBML — biophysical model translation and standardization Mar 2022 Latin Hypercube Sampling with Multi-Dimensional Uniformity (LHSMU) — multidimensional parameter sampling tool Jun 2020	
BOOK CHAPTERS	Bhatia, A., Moza, S., Bhalla, U.S., “Patterned Optogenetic Stimulation using a DMD-projector”, <i>Channelrhodopsin</i>, Chapter 11, Springer Protocols, 2020	
RESEARCH COMMENTARY / HIGHLIGHTS	Moza, S., “Action at a Distance: Theoretical Mechanisms of Cross-Dendritic Heterosynaptic Modification”, <i>eNeuro</i>, 2023	
SELECTED TALKS, CONFERENCES & WORKSHOPS	Neuroscience 2025, Society for Neuroscience (SfN), San Diego, USA Nov 2025 Analysis and Modeling of Connectomes, Janelia Research Campus Jun 2025 CeNeuro, Wisconsin, Madison, USA Jun 2024 Neuronal Circuits, CSHL, NY, USA Mar 2024 Harvard MCZ Seminar Series May 2023 NeuroMatch Conference, Online Webinar March 2020 No Garlands Neuroscience, IISER Pune, India Jan 2020 Transylvanian Experimental Neuroscience Summer School, Romania Jun 2019 Molecules and Memory, NCBS, Bengaluru, India Mar 2019 Quantitative approaches to Behaviour & Neural Systems, Lisbon, Portugal Oct 2018 Spikes lecture series, Centre for Neuroscience, IISc Bengaluru, India Jan 2018 Neuroscience 2017, Society for Neuroscience (SfN), Washington DC, USA Nov 2017 No Garlands Neuroscience, IISER Pune, India Oct 2017 BSSE Symposium, IISc Bengaluru, India Jan 2017 Molecular Mechanisms at the Synapse, Janelia Research Campus May 2016	
TEACHING AND MENTORSHIP	Teaching Assistant and Organization Scienspurs Introduction to Computational Neuroscience -I & II Winter 2023 Computational approaches to memory and plasticity (CAMP) Summer 2014-18 National Centre for Biological Sciences, Bangalore Boston Bangalore Biosciences Beginnings Neuroscience school Winter 2016 Harvard University, Cambridge National Centre for Biological Sciences, Bangalore	
MENTORED PROJECTS	Panos (Masters): Now PhD student at Max Planck	
PROFESSIONAL SERVICE	Reviewer: eNeuro (2023-), J. Comp. Neuro. (2025-), J. Biosciences (2022-) Co-reviewer for Neuron, eLife and Cell (with senior collaborators)	
AWARDS & FELLOWSHIPS	Fellowships Council of Scientific and Industrial Research (CSIR) Senior Research Fellowship (SRF), Biology Jul 2014 - Jul 2017 Junior Research Fellowship (JRF), Biology (All India Rank 36) Jul 2012 - Jul 2014	

DBT Bioinformatics National Certification (All India Rank 33)	Feb 2011
Jawaharlal Nehru University- Masters Fellowship	Aug 2010 - Jul 2012

Travel Awards

IBRO-PERC, The Brain Prize and FENS stipend	May 2019
Wellcome Trust Travel Award	Sep 2018
Infosys Travel Award, Infosys Foundation	Dec 2017
Department of Biotechnology Travel Award, Government of India	Nov 2017

WET LAB

Model organisms: *C. elegans* and *Drosophila melanogaster*
 Techniques: genetics, optogenetics, calcium imaging, behavior

COMPUTATIONAL SKILLS

Modeling: Large-scale simulations, dynamical systems, information theory, control-based system identification, chemical reaction network modeling, stochastic simulation (Langevin/Gillespie), optimization
Analysis: Expertise in processing high-dimensional, volumetric (4D) neural datasets, neural time series modeling, connectome topology, tensor decomposition (SVD/CP/Tucker), graph-based learning, unsupervised learning and clustering
Tools: Python, PyTorch, JAX, scikit-learn, Optuna, NetworkX, NumPy, pandas, [MOOSE](#), [CoPaSi](#), [Brian](#) **Systems:** UNIX, multiprocessing, SLURM/SGE cluster environments